Module 1: Fundamentals of Climate Change and Policy Context

Module Instructor: Dr. James Kaoga

Module duration: 1 week (12 hours)

Module Overview

Climate change is one of the critical global issues requiring a global consensus and action. International Environmental Law is emerging as a critical asset in addressing climate crisis. This module introduces the learner to fundamental concepts of climate change and the foundations of major Climate Change policies, treaties and legislation at subnational, national, regional and global levels. At the end of the module, the learner will have a basic understanding of the key climate change concepts, and the main international treaties and conventions.

Learning Objectives

The learner will be able to**:**

1. define key concepts and terminologies related to climate change
2. summarize historical and the present state of climate change polices
3. point out the process towards a treaty making process within the context of global climate change
4. summarize the main climate change policies and treaties
5. critique the main climate change polices and treaties

Expected Learning Outcomes

At the end of this course the learner should be able to: -

1. be familiar with key concepts and terminologies to related to climate change
2. demonstrate familiarity with the history of sustainable development
3. illustrate the process towards a treaty within making within the context global climate change
4. demonstrate familiarity with the main global policies and treaties governing climate change
5. Appraise the main climate change policies pointing out challenges and opportunities

Module content

1. Definition of key concepts and terminologies related to climate change
2. Sustainable development history and motivations of climate change policy
3. The process towards treaty making at global climate change
4. United Nation Framework on Climate Change (UNFCCC) and Kyoto Protocol
5. Convention on Biological Diversity
6. Convention to Combat Desertification
7. Paris Agreement
8. Synthesis of the main climate change policies and treaties

Core Reading Materials

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Topic 1: Introduction to climate change policies

Climate Change conventions, treaties, protocols and legislations aim at reducing risks and impacts of climate change. They involve global, regional, national and sub national instruments that control human behavior and activities associated with climate change.

Definition of key concepts and terminologies of climate change

Climate change: This is an alteration in the state of the climate that can be identified by changes in the mean and/or the variability of its properties and that persists for an extended period, typically decades or longer.

- Climate change may be due to natural internal processes or external forcings, or persistent anthropogenic human activities causing changes in the composition of the atmosphere
- United Nation Framework Convention on Climate Change (UNFCCC) defines climate change as a change of climate which is attributed directly or indirectly to human activity that alters the composition of the global atmosphere and which is in addition to natural climate variability observed over comparable time periods.

Adaptation: Means adjusting to any new climatic conditions caused by natural and human-induced change.

- Many parts of the world have had to cope with previous climate variability and have developed strategies to do so.
- Adaptation can be undertaken in anticipation of impacts or after the disaster. These can be initiated by individuals or through coordinated measures by government or other groups.
- The mix of anticipatory or reactive adaptation action actually introduced will depend on the existing vulnerabilities of each community or individual, as well as institutional processes, regulatory frameworks, property rights, access to resources and social conditions.

Mitigation: Is defined as a human intervention to reduce the sources or enhance the sinks of greenhouse gases, technological change and substitution that reduce resource inputs and emissions per unit of output.

- With respect to climate change, mitigation means implementing policies to reduce (Greenhouse gas) GHG emissions and enhance sinks.

Climate change policies: These are measures put in place by governments to ensure economic growth under adequate mitigating measures and safeguards against adverse effects to the environment

International treaty/ Convention/ protocol/ covenant/acts/ memorandum of understanding/ statute:

- This is a formal agreement between States, normally open for participation by the international community as a whole, or by a large number of States.
- This is a means of creating international rules or standards that states and other actors of international community are expected to abide by.
- This denote a 'merger of wills of two or more international subjects for the purpose of regulating their interests by international rules.

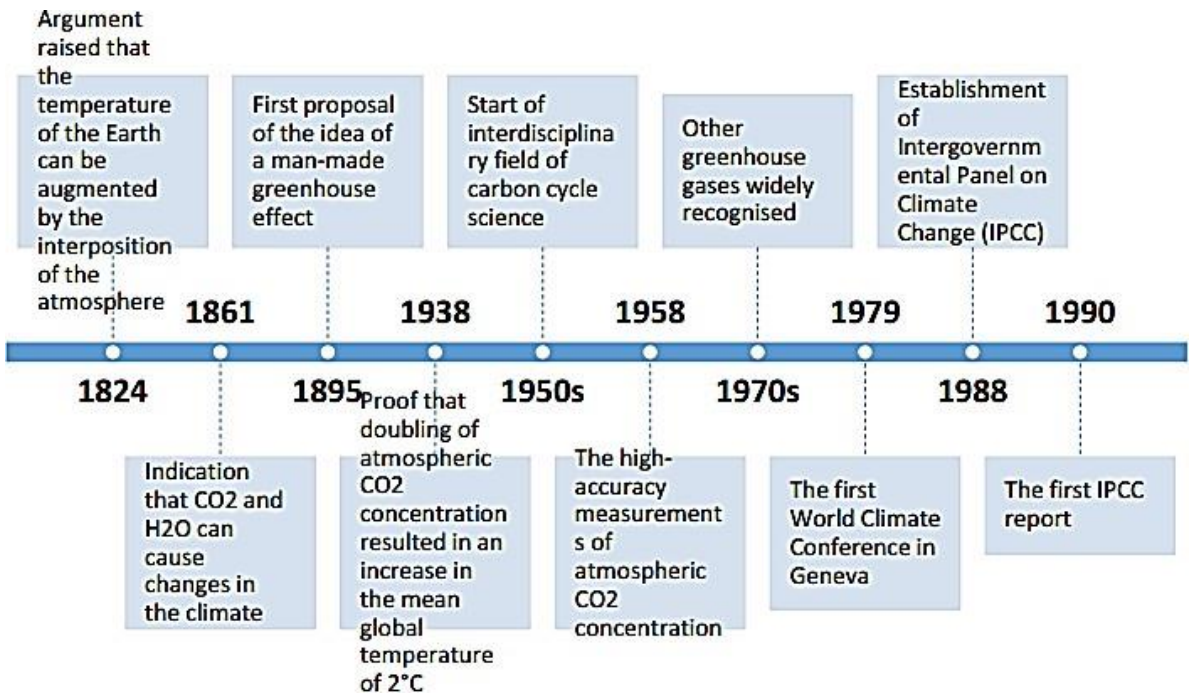

Figure 1: History of climate change science

The Fig 1 is an illustration showing a summary of history and motivation of climate change policies after the realization that the anthropogenic human activities amplify the pace and scale of climate change as described below:

1824: French physicist Joseph Fourier is the first to describe the Earth's natural "greenhouse effect".

1861: The Irish physicist John Tyndall shows that (carbon dioxide) CO_2 and (water) H_2O can cause changes in the climate

1895: Swedish chemist Svante Arrhenius concludes that industrial-age coal burning will enhance the natural greenhouse effect

1938: British engineer Guy Callendar shows that temperatures had risen over the previous century due to increased CO^2 concentrations.

1958: Geochemist Charles David Keeling was employed to continuously monitor CO_2 levels in the atmosphere, with an increase in Antarctica visible after only two years.

1970s: Other greenhouse gases from anthropogenic human activities: Nitrious Oxide (N_2O); Chloro Fluro Carbons (CFCs) Methane (CH_4)

1979: First World Climate Conference was held in Geneva, leading to the establishment of the World Climate Programme.

1988: Intergovernmental Panel on Climate Change (IPCC) is set up by the World Meteorological Organization (WMO) and the United Nations Environment Program (UNEP).

1990: IPCC delivered its First Assessment Report on the state of climate change, predicting an increase of 0.3 °C each decade in the 21st century.

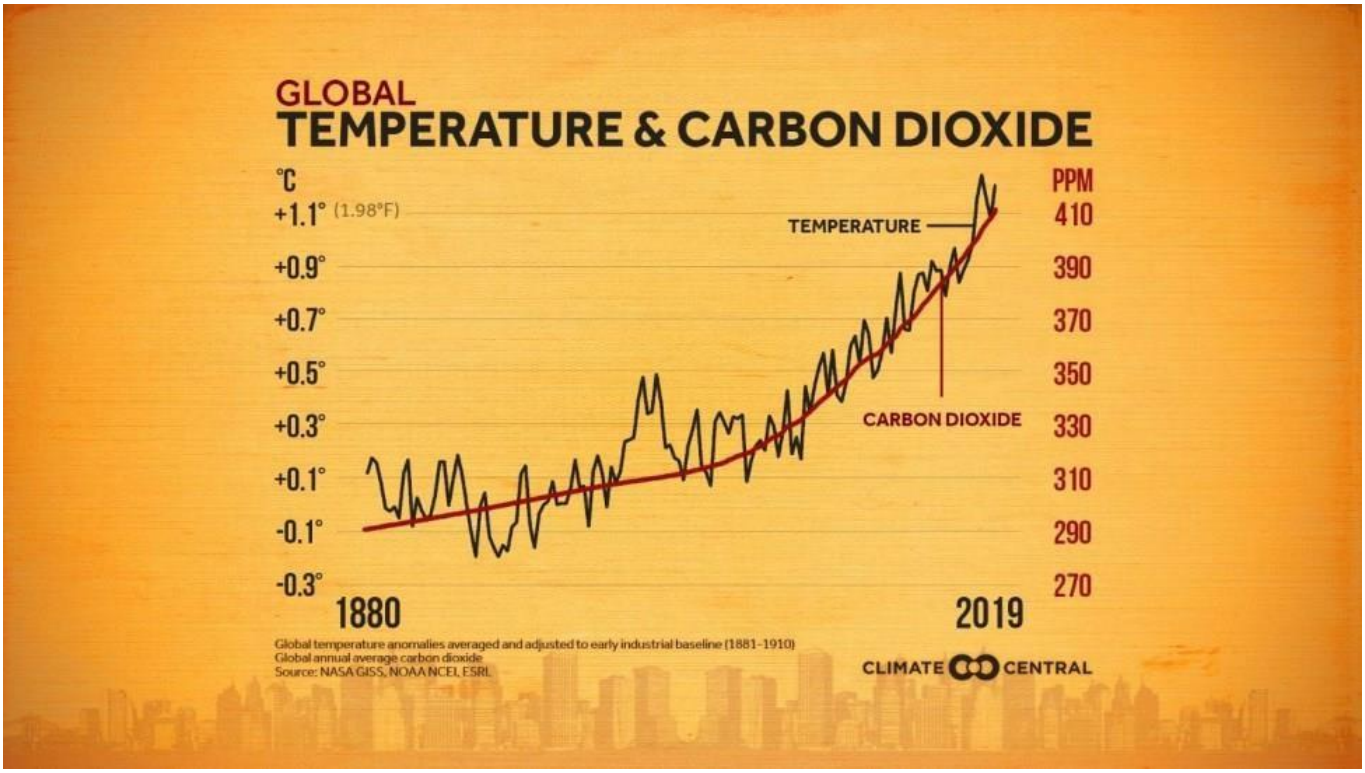

Figure 2: Correlation between CO_2 concentration and global warming

The Fig. 2 illustrate the correlation between CO_2 concentration is one of the many greenhouse gasses (GHG) that is behind global warming

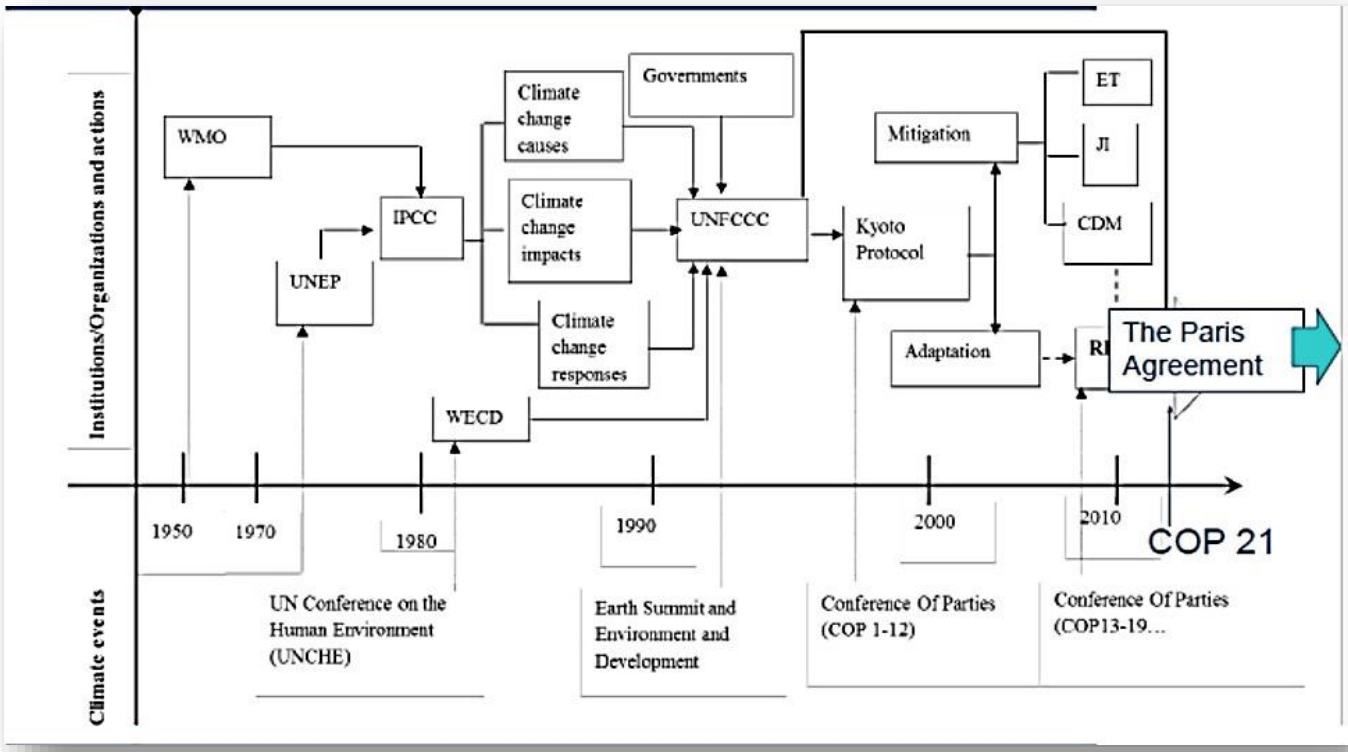

Figure 3: History of climate change policy making towards the Paris Agreement

The Fig 3 is an illustration showing history of climate policy making process towards the Paris Agreement:

1972: United Nations Conference on the Human Environment: Trigger global environmental action - Sustainable Development – 'development that meets the needs of the present without compromising the ability of future generations to meet their own needs' World Commission on Environment and Development (WCED, 1987:43).

Sustainable development: Entails development that meets the needs of today's generation without compromising those of the future generations

1988: Intergovernmental Panel on Climate Change

1992: UN Framework Convention in Climate Change signed

1997: Kyoto Protocol - is the first legally binding international climate agreement adopted at the 3rd Conference of Parties (COP) in Kyoto, Japan and commits industrialised nations to meet emissions reduction targets of 5.2% below 1990 levels during the 2008-2012 commitment period. The mechanism of implementation was largely market based and strictly legally regulated- Clean Development Mechanism (CDM), Joint Implementation (JI) and Emission Trading (ET).

2015: Paris Agreement (PA) -was enacted at COP 21 in Paris and aims to strengthen the global response to climate change by keeping a global temperature rise this century well below 2° degrees Celsius above pre-industrial levels and to pursue efforts to limit the temperature increase even further to 1.5° degrees Celsius.

Table 1: Major Convections and main objectives

Convections	Objectives
The Rio Declaration on Environment and Development	A set of principles which provide guidance on achieving sustainable development
UN Framework Convention on Climate Change	An international treaty to stabilize greenhouse gas concentrations in the atmosphere.
Convention on Biological diversity	An international convention with three objectives: the conservation of biodiversity, sustainable use of its components, and equitable sharing of benefits from the genetic resources
Convention to Combat Desertification	An international convention to combat desertification in those countries experiencing serious drought or desertification, particularly in Africa

UN Conference on Environment and Development (UNCED) also known as the Earth Summit, established a number of international agreements, declarations and commitments as shown in Table 1

Topic 2: The process towards a treaty development

Introduction:

Conference of Parties (COP) is a crucial platform where 197 signatories of UNFCCC converge annually to deliberate on progress being made to combat climate crises. In 2025, COP 30 was held in Belem, Brazil from November 10-21, 2025.

COP is a global platform where negotiations and policy development takes place. This notwithstanding it has been faced with coordination challenges especially striking the right balance at the local, regional and international levels in the harmonization of climate governance faced with institutional fragmentation, overlapping mandates, and conflicting national interests

Figure 4: Negotiating Groups

Fig 4 shows varied groups involved in the negotiation at the COP each of them pushing for their own interest.

COP is the supreme decision-making body of the convention and it takes place every year in a different country and region.

Most importantly, the climate change debate has become critically important for International communities, governments and individuals to address. Here are the major accomplishments of COP1 through COP30:

COP 1 (1995) The Inaugural Meeting – Berlin, Germany

Berlin held the opening session of the COP meetings in 1995. At COP1 the parties established fundamental procedures for the COP events including yearly meetings and a system where different countries would take turns holding the presidency. The Berlin Mandate emerged as a significant outcome from COP1 that required industrial countries to enhance their climate pollution reduction efforts. COP1 established essential framework guidelines for future climate agreements yet failed to establish direct emissions targets but it showed that developed nations had to provide financial support for climate change initiatives in developing countries. These foundations built upon the Rio Earth Summit of 1992 would enable significant progress within the climate system.

COP 2 (1996) Bridging Progress – Geneva, Switzerland

The conference happened at the same time as the Intergovernmental Panel on Climate Change released its Second Assessment Report combined with a World Health Organization summit which demonstrated climate change health effects. The main purpose of COP2 was to link the Berlin COP1 meetings with Kyoto COP3 preparations by providing parties a chance to collaborate on proposed future actions and debate their progress from COP1. Participants at COP2 maintained their focus on strengthening climate change commitments through renewed commitments that pushed forward the process of establishing binding targets. Parties maintained their focus on both Berlin Mandate talks and groundwork development which led to the creation of the Kyoto Protocol (Newell & Paterson, 1996).

COP 3 (1997) The Kyoto Protocol – Kyoto, Japan

The Kyoto Protocol emerged as the first significant binding international compact related to greenhouse gas emission reduction during COP3. The countries of developed nations committed

in Kyoto Protocol arrangements to lower their total emissions by around 5% relative to 1990 levels from 2008 until 2012. The Kyoto Protocol introduced specific emission reduction goals along with timeframes which established it as a groundbreaking achievement in worldwide climate management (Fletcher, 1998). Through the protocol developers established carbon market systems through emissions trading and the Clean Development Mechanism (CDM) which let developed countries support projects in developing countries for achieving emissions reductions. Through COP3 countries agreed to implement the Kyoto Protocol which established a new system for worldwide climate regulation while empowering international carbon market participation.

COP 4 (1998) The Buenos Aires Plan of Action – Buenos Aires, Argentina

The adoption of the Kyoto Protocol had just completed but had not entered into effect so parties focused on how implementation would operate. Parties reached an agreement at the conference which produced the Buenos Aires Plan of Action (BAPA) aimed at developing operational rules for the Kyoto Protocol over two years. The Buenos Aires Plan of Action concentrated on financial structures as well as technological transfer frameworks while defining specific timeframes for establishing flexible system regulations including emissions trading and CDM. COP4 served to advance the climate agenda through specifying operational steps needed to bring Kyoto Protocol into effect (UNFCCC, 1999). Subsequent meetings after this event concentrated on building operational rules and guidelines which would help implement Kyoto standards.

COP 5 (1999) Strengthening Commitments – Bonn, Germany

The conference functioned as a technical event to build stronger support for Kyoto Protocol implementation and sustain the process of making it operational. The delegates in Bonn refined draft decisions which had received previous conceptualization during Buenos Aires Plan of Action. The main purpose of COP5 included advancing discussions about complex implementation matters without concluding any new substantial agreements keeping it consistent with its follow-up nature. The conference at Bonn used focus on translating general Kyoto goals into specific protocols to retain member countries involvement thus continuing progress in global climate action. The talks at Bonn established necessary mechanisms for CDM enactment and compliance

systems through iterative discussion despite encountering implementation obstacles (UNFCCC, 2000).

COP 6 (2000) Challenges in The Hague – The Hague, Netherlands (resumed in Bonn, 2001)

COP6 became notable for the challenges in reaching agreement on Kyoto Protocol implementation. The talks at The Hague stalled amid significant disagreements between parties on issues like carbon sinks and financing mechanisms, leading the conference to suspend without a final deal. This impasse demonstrated the difficulties of balancing different national interests within a global accord (Huber, 2013).

COP 6-2 (2001): Resolving Disagreements - Bonn, Germany

To resolve the deadlock, an extraordinary resumed session known as COP6-2 took place in Bonn, Germany, in July 2001. At this follow-up meeting, negotiators successfully brokered compromises and finalized many operational details of the Kyoto Protocol (Huber, 2013). Key outcomes included agreement on rules for the CDM and other flexibility mechanisms, clearing the way for nations to move forward with ratification. The successful conclusion of COP6-2 showed the determination of the international community to keep the Kyoto process on track despite setbacks.

COP 7 (2001) Advancing Climate Action – Marrakesh, Morocco

Building on the progress from Bonn earlier that year, COP7 delivered a comprehensive set of decisions known as the Marrakesh Accords. These accords finalized the rulebook for implementing the Kyoto Protocol, including detailed guidelines for emissions trading, joint implementation, and the CDM, as well as compliance mechanisms and accounting methods. Importantly, COP7 also adopted the Marrakesh Ministerial Declaration, which highlighted the promotion of development (linking climate and sustainable development) and paved the way for future discussions on adaptation and support for least developed countries. Additionally, the conference established the first frameworks for National Adaptation Programmes of Action (NAPAs) to help the most vulnerable countries plan for climate resilience (NAP Global Network, 2020). The outcomes of Marrakesh provided the legal and technical backbone for the Kyoto Protocol's entry into force, solidifying many outstanding issues from previous COPs and advancing climate action into a new phase.

COP 8 (2002) Strengthening Kyoto Protocol – New Delhi, India

After the Kyoto rulebook received approval at COP8 the conference moved onto developing strategies for better climate initiatives along with supporting developing nations' requirements. The conference implemented every element of the Marrakesh Accords to bring the Kyoto Protocol into operation (Huber, 2013). The Delhi Ministerial Declaration on Climate Change and Sustainable Development received backing at COP8 as it stated the development requirements of impoverished countries while asserting that technology transfer should proceed jointly with adaptation alongside mitigation measures. The idea of establishing a specific climate fund originated at COP8 before becoming the operational Green Climate Fund (Huber, 2013). During COP8 the conference managers upheld the core concept that climate action must merge sustainable development with poverty elimination work.

COP 9 (2003) Outreach and Progress on CDM and Sustainable Development – Milan, Italy

The conference gained recognition as "forestry COP" because it successfully moved forward important technical climate framework aspects especially related to land-use and forestry matters (UNFCCC, 2003). Under the CDM participants created extensive guidelines which explained eligibility criteria for issuing forestry activity credits to countries. The COP 9 conference enhanced the Delhi Declaration's major concepts by promoting capacity development and technological assistance for developing nations to establish environmentally friendly development models. Sessions made significant operational progress in implementing the Kyoto mechanisms while handling regular matters of the Subsidiary Bodies (SBSTA and SBI) to advance implementation. The decisions made at COP 9 failed to generate prominent agreements but they enhanced the framework for future action through strengthened trust and cooperation (UNFCCC, 2004).

COP 10 (2004) Reviewing Achievements – Buenos Aires, Argentina

This conference analyzed an entire decade of climate change activities on its tenth anniversary of the UNFCCC introduction. The delegates examined climate programs initiated after 1995 to determine their effectiveness and challenges (Huber 2013). The Buenos Aires Programme of Work on Adaptation and Response Measures became a primary outcome of the conference by

intensifying worldwide focus on adaptation within climate policy. The program specified guidelines to assist countries during climate adaptation while including provisions for vulnerable groups affected by policy side effects (mainly regarding fossil fuel export economies). At COP 10 negotiators acknowledged that both adaptation and equity needed to gain equal standing with mitigation initiatives as a major shift in negotiations occurred. The lack of new binding emission targets during COP 10 led to a major shift in global climate discussion by making adaptation funding and technology transfer essential elements of the climate agenda (UNFCCC, 2005).

COP 11 (2005) Finalizing the Kyoto Protocol – Montreal, Canada

COP11 marked an important point in the climate process because of its dual accomplishments. The conference functioned as the very first Kyoto Protocol Meeting of the Parties (CMP1) since the protocol officially became effective in 2005. The conference completed the essential rulebook requirements of the Kyoto Protocol to enable its full operational implementation. The Kyoto Protocol gained executive readiness through the adoption of final operational procedures and compliance rules which COP11 coordinated with the Kyoto parties' meeting to provide nations with a solid legal framework to meet their emission goals (UNFCCC, 2005). Leaders introduced the Montreal Action Plan which started negotiations for long-term cooperative action spanning beyond 2012 which marked the termination of the first commitment period under Kyoto. The blueprint began the process to secure upcoming emission reduction commitments although the designers recognized future demands would necessitate extended reductions extending beyond Kyoto. The accomplishments of COP11 achieved two goals which propelled the international climate initiative by finalizing the Kyoto protocol structure while planning for post-2012 global action.

COP 12 (2006) Initiating New Commitments – Nairobi, Kenya

The climate regime received greater scope at COP 12 through adaptation measures. The Nairobi Work Programme on Adaptation served as a major outcome by establishing a thorough initiative which helps governments especially developing nations recognize climate impacts as well as develop appropriate responses. The Nairobi Work Programme operated as a platform for adaptation practice exchange and it provided crucial information to guide future funding decisions and planning decisions. The review of future obligations under the Kyoto Protocol started at COP12 and Belarus received inclusion in Annex B which added another contribution-focused

country to emission reduction requirements. Participants during COP 12 approved the principles that would enable operations of the Adaptation Fund through CDM revenue contributions (Ancha et al., 2006).

COP 13 (2007) The Bali Roadmap – Bali, Indonesia

The Bali Roadmap gained its importance following COP 13. An agreement replacement needed to be established because the first commitment term of the Kyoto Protocol was concluding. The Bali Roadmap determined how to create a future climate regime through a planned two-year negotiation schedule. The Bali Action Plan functioned as the core component of the roadmap since it defined essential blocks for the forthcoming agreement through mitigation together with contributions from developed and developing nations alongside adaptation measures and technology transfer and financial assistance aspects. The Bali Roadmap established negotiations that would lead to the Paris Agreement through 2009 negotiations (Christoff, 2008). The Bali Partnership dealt with two key issues during COP13 by achieving technical sharing agreements and creating methods for forest protection which laid ground for upcoming REDD+ frameworks. The Bali conference achieved its recognition through its ability to merge all negotiating parties from the United States to key developing nations behind a single objective of developing a thorough new climate treaty while illustrating joint international support for expanded climate change solutions.

COP 14 (2008) Bridging Progress Between Bali and Copenhagen – Poznań, Poland

The Poznań conference acted as a pivotal meeting between the vital Copenhagen meeting and its following year. The Poznań conference delegates strived to sustain progress between the Bali Roadmap and the upcoming Copenhagen summit. At COP 14 parties enabled the operational start of the Adaptation Fund through rule-making for accessing funds so developing countries could procure support for adaptation initiatives (UNFCCC, 2008). The conference did not produce any significant new policy achievements or emission commitments during COP 14. Countries were invited to keep building their national plans during the conference while the meeting primarily focused on reviewing existing activities. The Poznan conference received global approval for its objective to develop Copenhagen (COP15) by solving technical issues in advance. COP 14 established the need for continuous negotiation progress during bannerless pacts to assure future high-level conferences achieve their objectives.

COP 15 (2009) The Copenhagen Accord – Copenhagen, Denmark

The Copenhagen COP 15 emerged as the most expectant climate convention responsible for creating the Bali Roadmap's envisioned agreement. The negotiations in Copenhagen led to the creation of the Copenhagen Accord which served as a political agreement instead of becoming a formal decision through the UN. The COP adopted the Accord as a non-binding agreement after all countries except one failed to reach unanimous approval for it. The document established 2°C as the accepted maximum temperature increase threshold while obtaining voluntary emission reduction commitments both from developed countries and major developing nations. This agreement established that developed nations would provide new funds totaling approximately $30 billion during 2010–2012 followed by their commitment to create $100 billion yearly financing for developing countries by 2020 (Center for Climate and Energy Solutions, 2017a). Though non-binding the addition of finance provisions served as vital recognition toward fairness and support between countries during the Copenhagen summit. The Copenhagen Accord introduced a voluntary approach to climate pledges yet it established inclusive participation in climate negotiations even though it failed to obtain treaty status. The difficult experiences at COP15 shaped how subsequent major global climate change agreements approached their development.

COP 16 (2010) Formalizing Commitments – Cancún, Mexico

The Cancún Agreements from COP16 served to rebuild confidence in the United Nations climate process which had been damaged by Copenhagen by forming a comprehensive set of decisions. The agreements transformed many Copenhagen Accord elements into formal provisions of the UNFCCC structure (Maslin et al., 2023). Through COP16 the world witnessed the creation of the Green Climate Fund as a dedicated global fund which provides support to developing countries for emissions reduction and climate impact adaptation. The Cancún Adaptation Framework and Adaptation Committee received establishment at this point while the process for developing National Adaptation Plans commenced to assist vulnerable populations. The conference led all nations to endorse the international warming limit of 2°C and require members to submit their UNFCCC-based climate action plans that formalized Copenhagen agreements. The diplomats achieved consensus at the Cancún conference which resulted in a diplomatic achievement. The GCF along with other formal commitments established at COP16 prepared an

enduring climate framework that showed international climate collaboration was returning (Center for Climate and Energy Solutions, 2017b).

COP 17 (2011) Durban Platform and Binding Commitments – Durban, South Africa

The Durban Platform for Enhanced Action proved to be the chief accomplishment of the conference by establishing negotiations to create future climate agreements applicable to all member states (Center for Climate and Energy Solutions, 2017c). The implementation plan for the upcoming agreement became "a protocol, another legal instrument or an agreed outcome with legal force" which parties in Durban decided to adopt by the year 2015 before finalizing in the Paris Agreement. The decision to start a new commitment period of the Kyoto Protocol served as a major outcome despite the lack of participation by major polluters because Europe and Australia together with other countries supported continuing the Kyoto targets until 2020. Through Durban conference efforts resulted in operationalizing two key initiatives including the GCF and the Technology Mechanism. The establishment of a standardized dialogue among every nation including developed and developing nations marked COP17 as a landmark event. The step created awareness that the world would prepare a unified climate agreement that covered every nation in contrast to Kyoto system-based sector-based responsibilities.

COP 18 (2012) Amendments and Gender Equality – Doha, Qatar

COP18 operated by establishing the Doha Amendment to the Kyoto Protocol. The Doha Amendment established the second Kyoto Protocol period (2013–2020) according to Durban agreements through which European Union and Australian countries accepted emissions obligations while major polluters remained absent (Center for Climate and Energy Solutions, 2017d). The Doha Amendment transferred the emission trading system and CDM component of Kyoto to become applicable during the new time span. COP18 became the first annual Climate Change Conference to tackle gender equality as a major topic of its climate deliberation session which produced decisions for better female inclusion in climate decision-making and gender-based climate policy. The conference made progress on the future negotiations under the Durban Platform for a 2015 agreement while member countries contributed their proposals for its content.

The lack of a substantial leap in ambition at COP18 failed to produce but the meeting successfully maintained continuity by bridging promise periods thus pushing climate action forward. The event brought about a discreet transition towards addressing social aspects of climate policy (gender included) which added depth to the climate discourse for future COP meetings.

COP 19 (2013) Warsaw Mechanism for Loss and Damage – Warsaw, Poland

The major achievement of COP19 became the creation of the Warsaw International Mechanism for Loss and Damage. The mechanism aimed to address climate-related loss and damage suffered by vulnerable developing nations since some areas needed more than adaptation and mitigation alone to handle inevitable damages. Through the Warsaw Mechanism countries obtained access to coordinated support that included knowledge exchange to handle extreme weather events and rising sea levels and other slow-onset events. During COP19 the parties established the "Warsaw Framework for REDD+" to provide funding with methodological guidance aimed at forest and deforestation emission reductions (Center for Climate and Energy Solutions, 2017c). The authorities of Warsaw launched discussions that encouraged nations worldwide to develop their intended nationally determined contributions prior to the upcoming Paris Agreement. The COP19 meetings experienced intense tensions but still achieved delivery of the loss-and-damage mechanism while maintaining the process leading to Paris through stronger recognition of climate justice issues.

COP 20 (2014) Lima Call for Action – Lima, Peru

COP20 proceeded with creating the basic framework needed for the future global climate agreement that would decide its terms in 2015. All countries were required to submit their Intended Nationally Determined Contributions (INDCs) in advance through the Lima Call for Climate Action which finalized the conference at COP20. The step was crucial to obtain agreement commitments between developed and developing nations ahead of the Paris Agreement. INDCs needed to contain specific information according to the Lima Call for Climate Action to enhance mutual awareness and drive higher ambitions. Through COP 20 the process improved draft Paris Agreement components while enhancing provisions related to climate financial support and operational development initiatives (COP 20 outcomes and critiques, 2023). Lima's main goal was achieved as the meeting accomplished both registering nations to the universal participation framework and maintaining countries on a path toward a Paris climate agreement. The

establishment of trust through COP20 confirmed the readiness of parties to handle their climate responsibility commitments.

COP 21 (2015) The Paris Agreement – Paris, France

The Paris Agreement emerged from COP21 as the most substantial international climate policy agreement after 196 parties adopted it during the Paris conference. This historical treaty serves as a legally binding document of the UNFCCC which requires all member parties to execute climate action plans. The Agreement sets its main goal to maintain temperature rise below 2 degrees Celsius above pre-industrial times while striving to reduce temperature to 1.5 degrees Celsius. Every participating country must produce and revise its nationally determined contributions (NDCs) containing emission reduction goals as part of the Paris Agreement through a new bottom-up collaborative framework. Under this framework every nation must report its collective achievements through global stock takes conducted once every five years for improved transparency. The Paris Agreement succeeded due to joined-up support among a large number of countries and worldwide backing following the Copenhagen failures. The Paris Agreement functions as a groundbreaking framework which launched global communities on their mission to advance climate action ambitions (Maslin, Lang, & Harvey, 2023). The meeting at Paris marked the COP process' most impactful accomplishment through the creation of this agreement which followed years of diplomatic development starting from the first COP gathering.

COP 22 (2016) Implementing the Paris Agreement – Marrakesh, Morocco

COP22 with its official name as the "Action COP" the conference worked to create specific operational protocols for the Paris Agreement implementation while confirming worldwide climate change responsibility. The parties at Marrakesh moved forward with implementing the Paris Agreement rulebook through the development of reporting and effort assessment mechanisms which relied on standardized transparency frameworks (Center for Climate and Energy Solutions, 2017f). The Marrakesh Action Proclamation provided COP22 with a political statement which supported international climate action even after nations changed their policies. Climate finance and adaptation received significant attention at the conference through discussions to ensure the $100 billion yearly support target would be met along with continuous financial flow escalation. As part of COP22 the first meeting of Parties to the Paris Agreement referred to as CMA1 officially activated the Paris Agreement. within the UN framework. While COP22 did not

create new headline agreements, it succeeded in translating the optimism of Paris into concrete steps forward: countries began turning pledges into action plans, and the global community showed unity in pressing ahead with implementation efforts.

COP 23 (2017) Advancing the Paris Agreement – Bonn, Germany (Fiji Presidency)

The gathering of COP23 ran from November 6–17, 2017 under the leadership of Fiji as the president while Germany served as the meeting host. At the Bonn conference new advances sought to complete the Paris Agreement work program which started in Marrakesh. The new initiative at COP23 introduced the Talanoa Dialogue based on Fijian story sharing traditions that facilitated community consensus. The Talanoa Dialogue served as a mechanism for reviewing worldwide climate goal progress before new National Determined Contributions were released alongside increased national ambitions. COP23 implemented the Gender Action Plan which integrates women's empowerment and gender-equity into climate initiatives while establishing the Ocean Pathway Partnership to promote ocean climate system understanding (Obergassel et al., 2017). The combined progress of these initiatives together with persistent funding for adaptation as well as loss and damage made COP23 play an essential role in sustaining forward movement. The technical aspects dominated the Bonn conference yet its results created essential conditions that guided COP24 decisions at Katowice.

COP 24 (2018) Finalizing the Paris Rulebook – Katowice, Poland

COP24's primary achievement was the adoption of the **Katowice Climate Package**, informally known as the Paris Agreement "rulebook." This package of decisions provided detailed guidelines on how the Paris Agreement would be implemented moving forward. It covered transparency rules (how countries report and monitor their emissions and climate actions), common frameworks for updating and communicating NDCs, the mechanics of the global stocktake, and other operational details. By finalizing the rulebook, COP24 gave clarity and predictability to countries on what was expected of them, enabling the full implementation of the Paris accord (UPSC, 2024). Another focus in Katowice was the scientific context: delegates highlighted the findings of the IPCC Special Report on 1.5°C (released in 2018) to press for greater urgency. While negotiations in Katowice were challenging — especially on issues like finance and

differentiation — the agreed outcome was a comprehensive set of rules that put all countries on a more equal footing in terms of reporting and accountability. COP24's success in delivering the rulebook was seen as critical to making the Paris Agreement operational and effective in the years to come.

COP 25 (2019) Advancing Carbon Reduction Goals – Madrid, Spain

COP25 took place in Madrid, Spain (under Chile's presidency), after being relocated due to unrest in Chile. Branded the "Chile–Madrid Time for Action," COP25 was intended to resolve the last major piece of the Paris Agreement framework that Katowice had deferred: the rules for international carbon markets under Article 6 of the Paris Agreement (Lennan & Morgera, 2022). However, despite marathon negotiations, countries at COP25 could not reach consensus on Article 6 guidelines, postponing the decision to the next year. This impasse over carbon trading rules made COP25 a disappointment for many observers. Nonetheless, the conference did reinforce other areas: parties launched a new "Climate Ambition Alliance" of countries and non-state actors committed to enhancing their 2030 targets, and there were forward-looking discussions on strategies for achieving net-zero emissions by 2050. COP25 also addressed gaps in adaptation and loss-and-damage finance by strengthening the mandates of the Warsaw Mechanism and establishing the Santiago Network to link vulnerable countries with technical assistance. The difficulty in Madrid underscored the persistent challenges in reaching consensus on complex issues in a UN setting. By the close, the COP urged countries to "enhance ambition" in their NDCs by COP26 and left the critical carbon market issue for future resolution.

COP 26 (2021) Full Implementation of the Glasgow Pact – Glasgow, United Kingdom

After a one-year delay due to the COVID-19 pandemic, COP26 was held in Glasgow, UK, (in partnership with Italy). COP26 resulted in the Glasgow Climate Pact, a comprehensive political agreement that reaffirmed the goals of the Paris Agreement and went further in certain areas. Notably, the Glasgow Pact was the first COP decision to explicitly target fossil fuels: it called upon parties to accelerate "efforts towards the phasedown of unabated coal power and phase-out of inefficient fossil fuel subsidies" (Lennan & Morgera, 2022). It also urged countries to come forward with stronger 2030 targets by the end of 2022, recognizing that current NDCs are insufficient to meet the 1.5°C goal. Importantly, COP26 finally resolved the outstanding elements of the Paris Rulebook that COP25 had failed to conclude: consensus was reached on Article 6

mechanisms, establishing rules for international carbon markets and a new mechanism to replace the CDM. Agreement on common timeframes for NDCs and a transparency framework was also achieved. In terms of finance, the Glasgow conference pushed developed countries to at least double their adaptation finance by 2025 and initiated deliberations on a new collective finance goal beyond 2025. While many noted that the Glasgow outcomes still fell short of what science demands, COP26 did mark a significant step forward. It closed out the negotiation of implementation rules and shifted focus decisively toward implementation and increased ambition this decade (Lennan & Morgera, 2022). The Glasgow Climate Pact thus stands as a key achievement, energizing global climate efforts in the post-Paris era.

COP 27 (2022) Loss and Damage Fund – Sharm El-Sheikh, Egypt

COP27 expectations for this conference centered on implementation and addressing the needs of vulnerable countries, and the outcome delivered a breakthrough decision: the establishment of a Loss and Damage Fund. After years of calls by developing nations, COP27 agreed to create a dedicated fund to help vulnerable countries cope with losses and damages from climate impacts (such as extreme weather events and sea-level rise). The creation of this fund was widely hailed as a major achievement, as it represents a new pillar of the climate regime focused on climate justice and recovery. In other areas, COP27 made incremental progress. The conference reaffirmed the goal of limiting global warming to 1.5°C and launched a work program to scale up mitigation ambition and implementation in this critical decade. On finance, there were strong appeals for developed countries to fulfill their $100 billion annual commitment and to reform multilateral development banks to better mobilize climate finance (Freund, 2023). COP27 also saw new pledges toward the Adaptation Fund and continued discussions on a Global Goal on Adaptation. While it did not significantly advance mitigation targets, COP27 maintained global consensus on key objectives and, significantly, put loss and damage squarely on the agenda with a concrete funding mechanism. This acknowledged the realities faced by vulnerable nations and added a new dimension of solidarity to international climate action.

COP 28 (2023) Fossil Fuel Transition – Dubai, United Arab Emirates

A centerpiece of COP28 was the first Global Stock take under the Paris Agreement – a comprehensive assessment of the world's collective progress toward the agreement's long-term goals. This stock take, concluded at COP28, revealed the gap between current national

commitments and the emissions reductions needed to meet the 1.5°C and 2°C targets, thereby informing the next round of NDC updates. In response to the findings, COP28 encouraged countries to strengthen their 2030 targets by 2025 and to develop ambitious mid-century strategies aimed at net-zero emissions. Another major focus in Dubai was the energy transition. COP28's presidency emphasized the need to accelerate the shift from fossil fuels to renewable energy globally, and discussions centered on scaling up clean energy investment and deployment. There was also debate about language on phasing down fossil fuels, reflecting a growing push for clearer commitments on coal, oil, and gas. Additionally, COP28 continued efforts to reform climate finance – exploring innovative funding mechanisms and urging international financial institutions to play a larger role in mobilizing climate investments (Freund, 2023). Overall, COP28 underscored the importance of regular check-ins on progress and aimed to catalyze faster action by highlighting both shortcomings and opportunities in global climate efforts.

COP 29 (2024) Climate Finance Expansion – Baku, Azerbaijan

COP29 was poised to drive the climate agenda forward with a strong emphasis on financial and technological support for developing countries. A headline goal for COP29 was to significantly expand climate finance: parties are discussing plans to roughly triple the flow of funds to developing nations, aiming to mobilize on the order of $1.3 trillion in climate finance per year by 2035 (Arora, 2025). Such an increase is seen as crucial for enabling mitigation and adaptation projects at the scale required to meet global goals. In addition to finance, COP29 is expected to advance agreements on carbon markets and energy storage solutions, building on the foundations laid in previous years. By enhancing international carbon trading frameworks and cooperation on clean energy technologies, the conference seeks to further enable cost-effective emissions reductions worldwide. COP29 will also review the commitments and outcomes of the first Global Stock take, pushing countries to enhance their NDCs and long-term strategies in line with its findings. If successful, the outcomes from Baku would mark a major leap in international support for climate action especially by delivering resources to where they are needed most and set the tone for the next phase of the global climate effort.

COP 30 (2025)- Funding for Developing Nations- Belém, Brazil

COP30 marked a pivotal moment for accelerating Paris Agreement implementation amid high geopolitical tension. The conference, focused on the Amazon, resulted in a "Belém Package," emphasizing increased funding for developing nations, a transition away from fossil fuels, and enhanced inclusion of Indigenous peoples. Adaptation to climate change was a top priority built on the new global finance goal agreed at COP29 in Baku which aims to scale up support for developing countries. Its aim is to mobilise at least USD 300 billion per year by 2035 in finance from public sources, as part of a total of USD 1.3 trillion per year from all sources, public and private.

Towards a treaty…. (the process)?

This is a relatively slow-moving process, however, a critical is a step in an undoubtedly long term international policy framework formation.

Many of the most pressing environmental challenges facing humankind today are global in nature, for this reason, international environmental law is becoming increasingly important as an instrument in attempting to counter global environmental especially climate change.

(i) A threat to a **common good** is identified through scientific research, observation, etc.

(ii) **Evidence** including **costs of no-intervention**, **intervention scenarios**, etc. is presented to policy mechanisms (mainly UN)

(iii) **Committee** appointed to look into and advise within **a fixed time-frame** with **specific products**, including draft text for negotiation

(iv) Member countries of the UN invited to deliberate on the report, including commencement of negotiations on the text

(v) Text could be **adopted** as a resolution of the UN organisation fronting this negotiated text.

Enforceability of International Law:

Climate change presents difficult questions calling for collective policy response at different levels of governance. Legislations are either proactive or reactive. These legislations began from

the realization of the dangers posed by climate change requiring the efforts from both developing and developed countries to reduce various risks of climate change.

- Under the Vienna Convention on the law of treaties 1969, the consent of a state to be bound by a treaty (and therefore for the treaty to apply to the state at an international plane) may be expressed by way of "signature, exchange of instruments constituting a treaty, ratification, acceptance, approval or accession or by any other means if so agreed"
- The Vienna Convention on the Law of Treaties 1969 was established to regulate treaty making between state parties as well as state parties and organisations.

Topic 3: International Climate Change Policies and Commitments

This topic introduces the learners to the main international climate change committees and commitments.

United Nations Framework Convention on Climate Change (UNFCCC)

- The UN Framework Convention on Climate Change (UNFCCC) is the main international agreement on climate change policy.
- It was commissioned in 1992 in order to consider what would be done to reduce the average global temperature rise and resultant climate change.
- Today, it has near-universal membership
- The 198 countries that have ratified the Convention are called Parties to the Convention
- The convention sets an overall framework for intergovernmental efforts to tackle the challenge posed by climate change.
- It recognizes that the climate system is a shared resource whose stability can be affected by industrial and other emissions of carbon dioxide and other greenhouse gases.
- The convention enjoys near universal membership, with 189 countries having ratified. It entered into force on 21st March 1994. Under the convention, signatory governments committed to:
    - Gather and share information on greenhouse gas emissions, national policies and best practices

- Launch national strategies for addressing greenhouse gas emissions and adapting to expected impacts, including the provision of financial and technological support to developing countries
- Cooperate in preparing for adaptation to the impacts of climate change.

Objectives of UNFCCC

- To stabilize greenhouse gas concentration in the atmosphere to a level that would prevent dangerous anthropogenic interference with the climate system, within a time-frame sufficient to allow ecosystems to adapt naturally to climate change;
- To ensure that food production is not threatened; and
- To enable economic development to proceed in a sustainable manner.

Structure of the UNFCCC

a. The Convention divides parties into three categories, which determine the obligations of the parties. The categorization essentially divides parties into developed and developing nations.
b. UNFCCC works with a secretariat to support all conventional bodies involved in international climate change negotiations; including: -
    i. Conference of the Parties (COP)
    ii. Conference of Meeting of the parties (CMP)
    iii. Subsidiary bodies which advise COP/CMP
    iv. COP/CMP Bureau-which deals with procedural and organizational issues arising from COP/CMP
c. The primary task of the conference of parties (COP) is the review the implementation of the convention. In particular, COP reviews:
    i. Scientific
    ii. Technical and technological advice
    iii. National reports on the progress of implementation
    iv. Adopts protocols, amendments and annexes as advised

Kyoto Protocol

- Kyoto Protocol is a global agreement on the reduction of the main greenhouse gas emissions that are linked to climate change and global warming.
- The Kyoto Protocol is adapted from the UNFCCC, agreed upon in 1997 and fully ratified and entered into force until February 2005.
- The Kyoto agreement places legally binding limits on emissions from developed countries in an effort to reduce climate change by 5.2% below 1990 levels, and unites.
- Kyoto Protocol provides for emissions trading and joint implementation of the target requirements (applicable only to Annex 1 countries)
- Developed countries must set national standards for estimating emissions directly, and achieving compliance with and enforcement of these emissions targets. Developed countries may also use ‘‘certified emissions reductions’’ from project activities in developing countries as credit towards GHG targets.

Convention on Biological Diversity

The text for this Convention (UNCBD) was adopted in Nairobi in 1992 and opened for signature at the Earth Summit in Rio de Janeiro in June 1992. It entered into force on 29 December 1993. Its objectives are:

- to ensure the conservation of biological diversity and the sustainable use of its components; and

- to promote a fair and equitable sharing of the benefits arising out of the utilization of genetic resources, including by appropriate access to genetic resources and by appropriate transfer of relevant technologies (taking into account all rights over those resources and to technologies), and by appropriate funding.

The biggest problem for negotiators over the UNCBD was how to justify restrictions of State sovereignty with respect to biodiversity conservation. The idea of biodiversity being a common heritage of humankind was seen, *inter alia*, as being an infringement of state sovereignty, so another approach was needed. What was decided upon was that the conservation of biological diversity is a 'common concern of humankind' and that biological resources are subject to State sovereignty.

The goals of the Convention are sought to be achieved through three broad strategies:

- promoting biodiversity conservation and sustainable use through national law and policy;
- creating an international institutional structure and policy framework to support implementation and achievement of the Convention's objectives; and
- Establishing a set of principles for the international exchange of genetic resources and the biotechnologies derived from them.

As far as the first strategy is concerned, the UNCBD's requirements are relatively flexible, but what is required is the development of a national strategy, plan or programme for the conservation and sustainable use of biological diversity and to integrate this into relevant sectoral and cross-sectoral plans. Several biodiversity programmes flow from this policy, including the drafting of a Biodiversity Bill. Articles 8-10 of the Convention specify certain types of conservation measures which are supposed to be given effect to in the national policy (e.g. conservation in protected areas).

As far as regulation of the biotechnology trade is concerned, there are several principles incorporated in the Convention which, although somewhat vague, do address the issue. Countries are required to 'endeavor to create conditions to facilitate access to genetic resources for environmentally sound uses by other Contracting Parties and not to impose restrictions that run counter to the objectives of this Convention'. Countries obtaining such resources from other countries must do so on 'mutually agreed terms' and after 'obtaining prior informed consent'. Moreover, buyers and source countries must arrange for 'fair sharing' of the benefits derived from the resources (ie compensation). Finally, the Convention requires parties to 'provide and/or

facilitate' transfer to other parties of biotechnology derived from genetic resources, on favourable terms if to developing countries.

The third strategy involves the creation of an international infrastructure for implementation of the Convention. To this end, the Convention establishes four bodies: a financial mechanism, Conference of the Parties, a Secretariat and a scientific advisory group.

The Cartagena Protocol on Bio-safety which was adopted in Montreal, Canada advocates for the safe transfer, handling and use of living modified organisms (LMOs) that may have an adverse effect on biodiversity with specific focus on trans-boundary movements. The protocol establishes an advance informed agreement procedure for imports of LMOs, incorporates the precautionary principle and details information and documentation requirements.

Convention to Combat Desertification

The United Nation Convention to Combat Desertification (UNCCD) was adopted in Paris on 17 June 1994 and comes into effect on 29 December 1996. The objective of the CCD is to combat desertification and mitigate the effects of drought in countries experiencing serious drought and/or desertification, particularly in Africa, through effective action at all levels, supported by international co-operation and partnership arrangements, in the framework of an integrated approach which is consistent with *Agenda 21*, with a view to contributing to the achievements of sustainable development in affected areas.

The CCD, rather than emphasizing State actions, focuses on process rather than substance and ensuring public participation in the development and implementation of plans to combat desertification - **a 'bottom-up' approach**. Obligations on parties include prioritizing the combating of desertification and drought, as well as the underlying socio-economic causes of these problems. At the heart of the CCD are the National Action Programmes, which provide the blueprint for implementation of the Convention.

COP 21 (2015) The Paris Agreement – Paris, France

The Paris Agreement emerged from COP 21 as the most substantial international climate policy agreement after 196 parties adopted it during the Paris conference. This historical treaty serves as a legally binding document of the UNFCCC which requires all member parties to execute climate action plans. The Agreement sets its main goal to maintain temperature rise below 2° degrees Celsius above pre-industrial times while striving to reduce temperature to 1.5° degrees Celsius. Every participating country must produce and revise its nationally determined contributions (NDCs) containing emission reduction goals as part of the Paris Agreement through a new bottom-up collaborative framework. Under this framework every nation must report its collective achievements through global stock takes conducted once every five years for improved transparency. The Paris Agreement succeeded due to joined-up support among a large number of countries and worldwide backing following the Copenhagen failures. The Paris Agreement functions as a groundbreaking framework which launched global communities on their mission to advance climate action ambitions (Maslin, Lang, & Harvey, 2023). The meeting at Paris marked the COP process' most impactful accomplishment through the creation of this agreement which followed years of diplomatic development starting from the first COP gathering.

Paris Agreement

The agreement's objective was to limit global warming to well below 2°C, preferably to 1.5°C, through national actions to reduce emissions, with clear and ambitious targets (Rogelj et al, 2016).

Principle of the Paris Agreement

Common but differentiated responsibilities and respective capabilities, in the light of different national circumstances

The 2015 Paris Agreement – Article 7

Established the global goal on adaptation; enhancing adaptive capacity; strengthening resilience and reducing vulnerability to climate change

Topic 4: Synthesis of the main climate change related protocols and their contribution to climate adaptation and mitigation:

The Conference of the Parties (COP) is the decision-making body of the United Nations Framework Convention on Climate Change (UNFCCC). The COP meetings serve as global platforms where nations get to negotiate climate policies and advance strategies to tackle climate change. Here we examine the achievements of COP 1 to COP 30, highlighting key policy decisions, agreements, and mechanisms introduced in each session.

The COP meets every year, unless the Parties decide otherwise. The first COP meeting was held in Berlin, Germany in March, 1995. The COP meets in Bonn, the seat of the secretariat, unless a Party offers to host the session. Just as the COP Presidency rotates among the five recognized UN regions - that is, Africa, Asia, Latin America and the Caribbean, Central and Eastern Europe and Western Europe and Others – there is a tendency for the venue of the COP to also shift among these groups.

The outcomes of COP 1 to COP 30 significantly shaped global climate policy, setting the stage for the historic Paris Agreement in 2015. The early COPs was crucial for establishing framework for international climate action. These conferences reinforced commitments to climate finance, adaptation, mitigation, and loss and damage mechanisms. While some COP meetings faced setbacks, their cumulative impact contributed to strengthening international climate governance and fostering long-term global cooperation.

The outcomes of COP 1 to COP 30 significantly shaped global climate policy, from the inaugural meeting in Berlin, Germany to Belem, Brazil the Conference of the Parties has continually shaped and reshaped international climate policy.

The early COPs were crucial for establishing framework for international climate action. Over the years, the COP process has produced groundbreaking agreements and mechanisms that form the core architecture of today's global climate governance.

Each COP added a building block: early meetings established foundational principles and institutions; the late 1990s and 2000s brought legally binding targets and innovative mechanisms under Kyoto;

The 2010s delivered the landmark Paris Agreement and an accompanying framework for transparency and review; and recent conferences have increasingly focused on implementation, finance, and addressing loss and damage to support vulnerable nations.

The steady progress of these negotiations, while often slow and challenging, reflects a growing international consensus on the need for collective action.

A key lesson from the COP history is the importance of scaling up ambition and cooperation over time.

As scientific urgency has grown, parties have had to continually increase their commitments moving from stabilizing emissions to actively cutting them, and now toward net-zero trajectories.

Equally crucial has been scaling up support: developed countries' financial pledges and technology transfers have enabled broader participation, and new instruments like adaptation funds and the Loss and Damage Fund address equity and justice.

The achievements from COP1 through COP 30 demonstrate significant strides in confronting climate change, yet they also highlight the work still ahead. To achieve a sustainable and resilient future for all nations, the world must build on these foundations with even stronger cooperation and action in the coming years. The COP meetings remain an indispensable forum for forging consensus and driving the global effort to combat climate change.

While some COP meetings faced setbacks, their cumulative impact contributed to strengthening international climate governance and fostering long-term global cooperation.

lack of participation by the United States, Australia and China hampers the unified ideal of global collaboration to solve the global problem of climate change.

One of the major problems is the conflict between countries. Conference of the Parties are treaty signed by different countries. When there is conflict among countries will lead to disagreements in the annual meeting, such as in the COP 27 there was great impact of the Russia and Ukraine conflict in the negotiations and agreements to be made. (Pandit, 2022). Secondly, the increase in the cost of living which happens globally, making some country become more vulnerable to climate change with time due to depletion of resources, shifting of the technology to be applied in the climate change. Thirdly, the problem of the consensus decision making especially, on the policies agreements and even treaties to be signed among countries like in COP 17. Lastly the

dependence of some countries on others and some they may delay the funds inform of grants and other loans to be issued.